Harikarthik A

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Education

Prince Shri Venkateshwara Padmavathy Engineering College **2022 - 2026**
Bachelor of Science in Computer Science(CGPA: 8.2 / 10) *Chennai,Tamilnadu*

Experience

Artificial Intelligence Intern **Nov 2025 – Jan 2026**
Infosys Springboard *Remote*

- Collaborated with team members to develop an AI-based Cat Emotion Recognition System using Python and deep learning, implementing a MobileNetV2-based CNN for facial emotion classification and an audio classification model for feline vocalizations. Contributed to image preprocessing, data augmentation, and MFCC-based audio feature extraction using Librosa for model training and evaluation.
- Worked collaboratively to integrate trained emotion recognition models into a Flask-based web application, implementing dedicated REST endpoints for image and audio predictions. Contributed to file upload handling, input validation, and prediction pipeline testing to support real-time emotion classification across multiple emotional categories.

Projects

AutoEval – Multi-Agent Developer Assessment Platform | *Python, FastAPI, LangGraph, Groq, pgvector, React, Docker*

- Architected a multi-agent AI pipeline using LangGraph that orchestrates dual LLM scoring (Llama 3.3-70B and Llama 3.1-8B via Groq) across 8 evaluation rubrics with a consensus layer that detects score conflicts greater than 1.5 points.
- Built a GitHub REST API ingestion layer and generated 384-dimensional semantic embeddings using sentence-transformers, stored in PostgreSQL with pgvector for semantic candidate similarity search.
- Implemented smart caching logic in the FastAPI backend to detect meaningful GitHub profile changes before triggering re-evaluation, reducing unnecessary score drift and API usage.
- Containerized the stack with Docker and deployed FastAPI services on Railway and Render with cloud PostgreSQL.

Cat Emotion Detection (Infosys Springboard Intern) | *Python, OpenCV, CNN, TensorFlow*

- Developed an AI-based system to detect cat emotions using image classification techniques.
- Built and trained a Convolutional Neural Network (CNN) model on labeled cat facial expression datasets.
- Used OpenCV for image preprocessing and feature extraction to improve model accuracy.

User Session Management App for 5G Networks | *Flask, Open5GS, UERANSIM, HTML, CSS, JavaScript, Shell*

- Developed a web-based application to automate UE and gNodeB operations in a 5G network test environment.
- Integrated Open5GS and UERANSIM with Flask backend to manage user sessions and monitor network nodes.
- Implemented shell scripts to automate starting and terminating UE and gNB sessions.
- Built a real-time dashboard using HTML, CSS, and JavaScript for monitoring logs and active nodes.

Technical Skills

Programming Languages: Python
Frameworks: Flask
Databases: MySQL
Web Technologies: HTML, CSS, JavaScript
APIs & Tools: REST APIs, JSON, Postman
Version Control: Git, GitHub
Deployment Platforms: Railway, Render, Vercel
Concepts: OOP, Machine Learning, Generative AI, RAG, Multi-Agent Systems

Achievements & Certifications

- **LEAP Program Winner – IITM Pravartak** – Built an Arduino-based smart heater with auto control.
- **AI Agent Hackathon** – Developed an NLP-based Q&A assistant.
- **TechImpact – GDG Hackathon** – Built a solution in a team-based hackathon.
- **AWS ML Foundations** – Learned core ML concepts and AWS basics.
- **Generative AI Prompt Engineering – Nunnari Labs** – Learned GENAI concepts and LLM-based applications.
- **5G Networks – Vaan Megam** – Learned Core of 5G Wireless Networks and Toplogies.